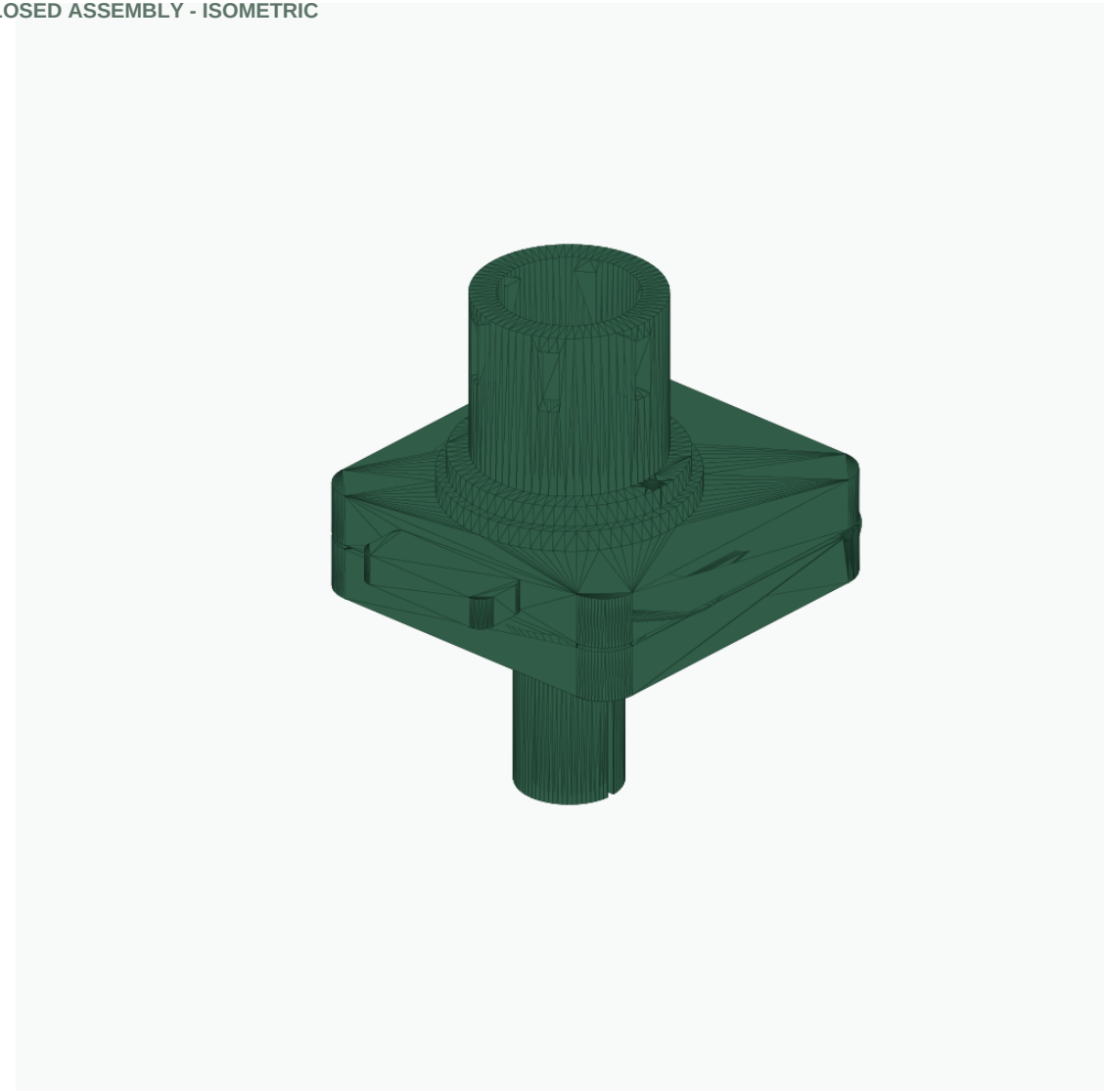
Prototype A-1: CAD design review

CLOSED ASSEMBLY - ISOMETRIC



Design intent

A compact, light-tight transmission clamp for the HYD1 fibre spectrometer. A controlled lamp illuminates a fixed leaf area; transmitted light is collected through an SMA905-compatible collimator beneath the leaf. Magnetic closure and four hard stops make the leaf spacing repeatable between operators.

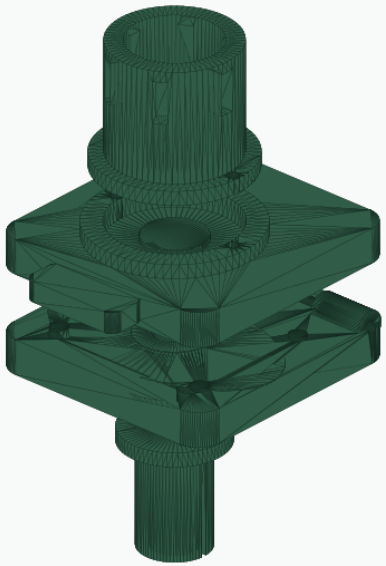
Envelope	70 W x 83 D x 99 H
Optical aperture	10 diameter
Leaf gap	0.8 nominal
Jaw plates	10 thick
Hinge pin	3 diameter
Closure	2 magnet pairs + hard stops
Source	Removable MR8 cartridge
Collector	Removable SMA905 cartridge

REVIEW STATUS

Concept geometry only. Lamp, diffuser, collimator and magnet interfaces remain provisional until purchased parts are measured.

Assembly and component breakdown

EXPLODED ASSEMBLY



1	**Source cartridge**	MR8 lamp, vented housing, diffuser outlet
2	**Upper jaw**	Lamp socket, upper gasket, magnets, centre hinge
3	**Leaf interface**	Two soft rings surrounding 10 mm aperture
4	**Lower jaw**	Hard stops, lower gasket, collector socket
5	**Collector cartridge**	Adjustable 12.7 mm collimator bore and clamp
6	**Hinge pin**	3 mm removable metal pin preferred

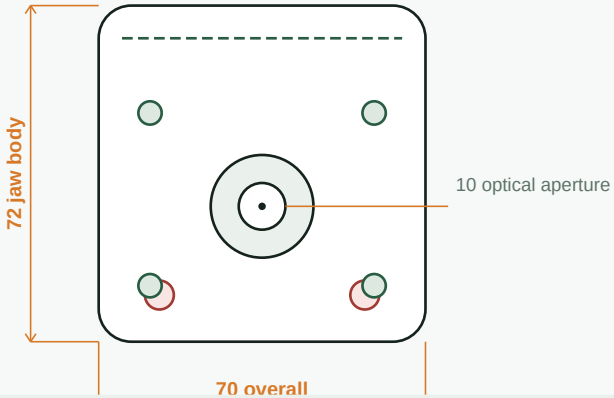
Assembly principles

- Optical cartridges remove without splitting the jaws.
- Gaskets are consumable and should be laser-cut or die-cut.
- Hard stops, not magnets, determine the closed spacing.
- Leaf contacts must be matte black and non-marking.
- Fibre bend radius is managed outside the clamp in A-1.

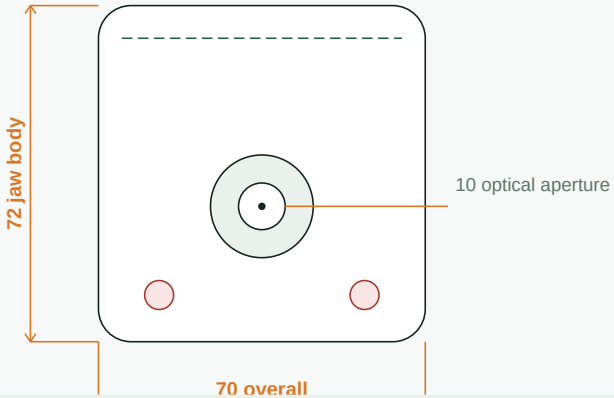
Jaw geometry

Plan views and locating dimensions

LOWER JAW - LEAF SIDE



UPPER JAW - LEAF SIDE

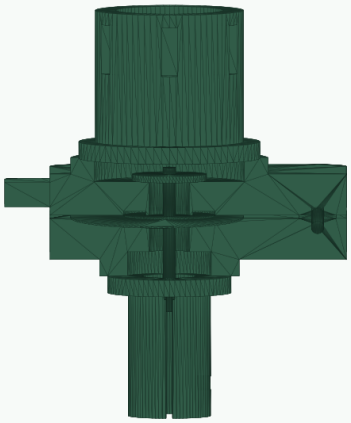


Datum: jaw centre. Optical centre is 7 mm toward the insertion edge. Magnet centres are x = +/-22, y = -26. Hinge axis is y = +29. Lower hard-stop centres are x = +/-24 at y = -24 and +13.

Optical section and interfaces

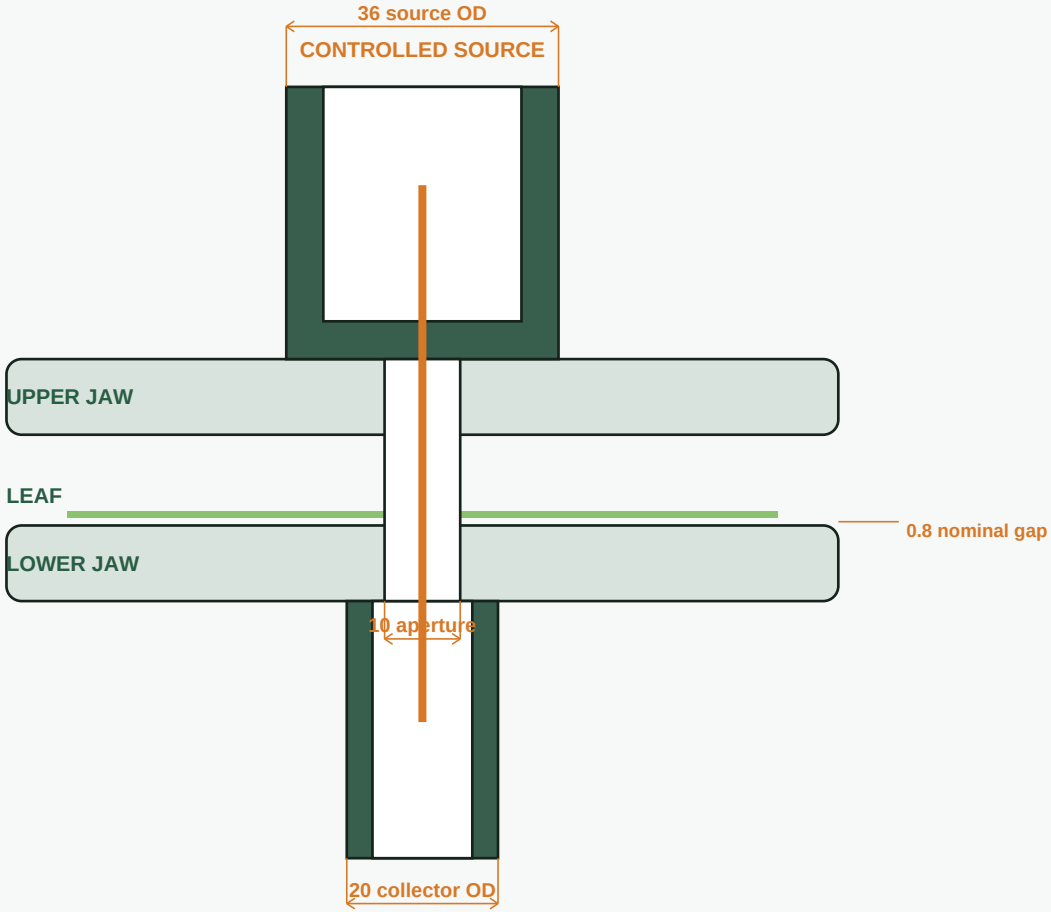
Not to scale - dimensional section

CAD CUTAWAY



Optical sequence

MR8 source -> diffuser -> 10 mm field aperture -> leaf -> 10 mm collection aperture -> quartz collimator -> SMA905 fibre -> HYD1.



Dimensions requiring purchased-part confirmation

Interface	R1 assumption	Required information
MR8 lamp	25.4 diameter	Actual reflector diameter, length, holder and lead exit
Diffuser	18 x 1.5	Material, clear aperture and mounting method
Collimator	12.7 diameter	Body diameter, length, focus adjustment and lock screw
SMA905 fibre	600-1000 um core	Connector body length and minimum bend radius
Magnets	6 x 3 discs	Grade, pull force and corrosion coating
Leaf gasket	22 OD / 12 ID / 0.8	Silicone hardness and compression behaviour

Proposed print specification

Black PETG or ASA; 0.20 mm layers; four perimeters; 35% gyroid infill; heat-set M3 inserts; print cartridges vertically; print jaws flat with leaf faces upward; paint or flock the optical cavities matte black.

R1 REVIEW QUESTIONS

1. Is a 70 x 72 mm jaw acceptable?
2. Should fibre exit vertically or be redirected toward the handle?
3. Do you prefer magnetic closure or a mechanical latch?
4. Should the worker hold the clamp or mount it on the main enclosure?